CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

Department Of Computer Science And Engineering

FUNDAMENTALS OF DATA SCIENCE (22CSC59)



CASE STUDY REPORT

Title: East-West Airlines: Customer Segmentation with Clustering

Submitted by:

Name: Billa Srishanth Reddy
Roll Number: 160124733166
Section: CSE-3
Year: B.E – 3 Year
Semester: V

Submitted to:

Faculty Name: Smt. S.Durga Devi

Academic Year: 2026–2027

1. Problem statement and objective

East-West Airlines seeks to understand customer differences in frequent-flyer activity, reward earning and use, credit-card miles, balances, and recent flight behavior. The objective is to identify useful customer segments using unsupervised clustering and interpret what distinguishes each segment. The supplied workbook contains the customer records and a description sheet noting that the company name was changed.

Dataset audit

Item Finding

Customer records 3,999

Columns in data sheet 12

Missing cells 0

Exact duplicate rows 0

Duplicate customer IDs 0

Clustering inputs 10 behavioral variables; ID# excluded; Award? reserved for profiling

Methodology

- Excluded ID# because identifiers do not describe behavior.

defining the segments directly by redemption status; it is used later to profile the segments.

- Applied log1p to highly skewed balance, qualification-mile, bonus-mile, transaction, recent-flight, and

enrollment-duration variables.

- Capped transformed skewed variables at their 1st and 99th percentiles, median-imputed missing values (none were

present), then used RobustScaler for distance-based methods.

- Compared K-Means for k = 2-8, hierarchical linkage variants, and DBSCAN across multiple eps/min_samples

settings. Evaluated with silhouette and other internal metrics where applicable.

Preprocessing rationale. Extreme frequent-flyer customers may be legitimate high-value customers. Capping reduces

their distance leverage without deleting their records; the trade-off is that the most extreme differences are

compressed. Results can change with alternative scaling or feature choices.

2. Statistical findings, EDA, and K-Means selection

Figure 1. Balance and bonus-mile distributions after log1p transformation for visualization. Both remain concentrated toward lower

values with a long upper tail, supporting the use of a skew treatment and robust scaling before distance calculations.

K-Means parameter evaluation

k Inertia Silhouette Davies-Bouldin Cluster sizes

2 18962 0.659 0.561 3773, 226

3 13164 0.360 0.939 2857, 916, 226

4 9761 0.354 0.998 719, 1973, 226, 1081

5 7980 0.330 0.969 724, 1246, 466, 1337, 226

6 7251 0.328 1.095 1179, 1065, 226, 467, 563, 499

7 6704 0.304 1.227 883, 464, 226, 439, 939, 577, 471

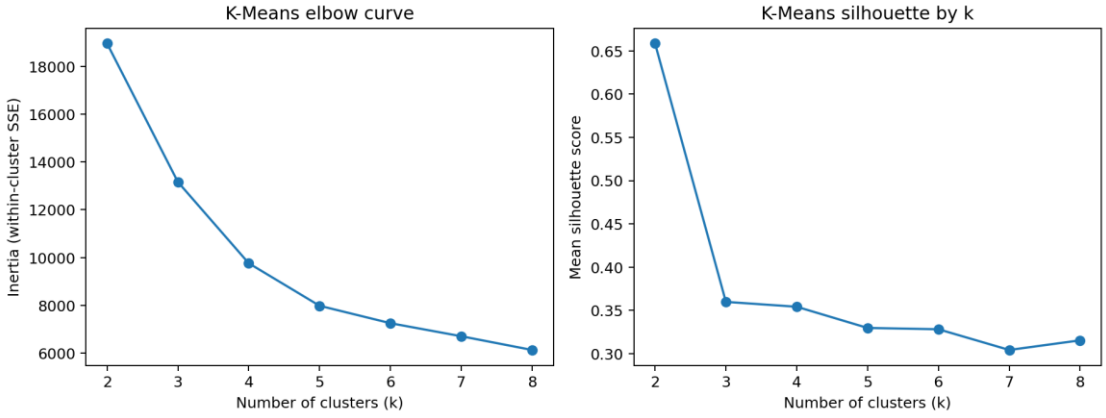
8 6130 0.315 1.113 815, 226, 429, 463, 874, 512, 230, 450

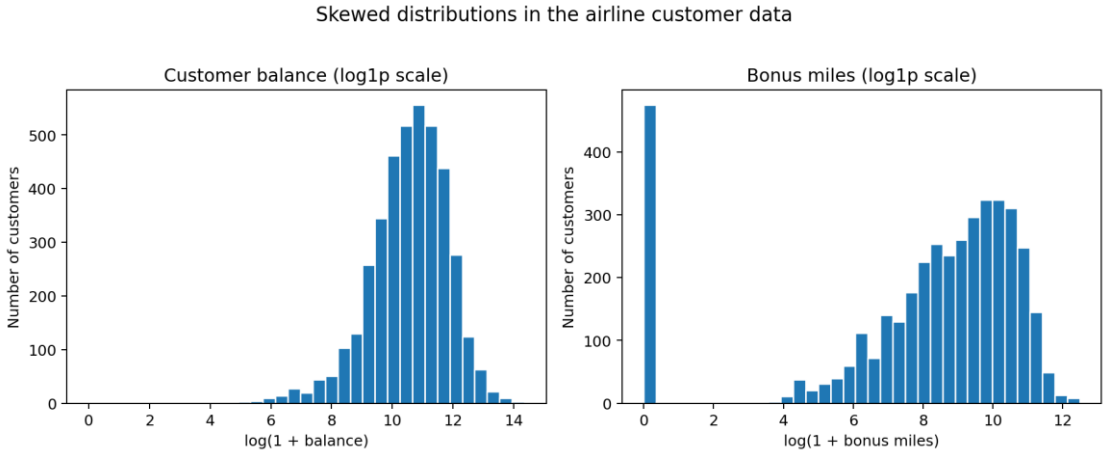
Figure 2. Inertia decreases as k rises, as expected; the silhouette score is highest at k = 2 (0.659). A two-cluster solution isolates the

226 customers with stronger recent flight activity from the much larger remainder. k = 3 is used as the operational profile because it

separates an additional low-reward-activity segment, but its lower silhouette (0.360) means its structure is less clean and should be

validated with the business team.





3. Model evaluation: Hierarchical clustering and DBSCAN

Hierarchical clustering was evaluated with Ward, complete, and average linkage at k = 3 and k = 4. DBSCAN was

tested using eps values from 0.75 to 2.50 and min_samples values of 5 and 10. For DBSCAN, silhouette is computed

on non-noise records only; noise points are not treated as one coherent cluster.

Selected algorithm comparison

Algorithm Clusters Silhouette Davies-Bouldin Noise

K-Means (k=3) 3 0.360 0.939 0.00%

Hierarchical Ward (k=3) 3 0.433 0.692 0.00%

Hierarchical complete (k=3) 3 0.639 0.709 0.00%

DBSCAN (eps=1.5, min_samples=10) 3 0.639 - 0.50%

Ward linkage provides a balanced three-cluster structure (silhouette 0.433). Complete linkage scores higher (0.639),

but isolates a very small 15-customer group, making the segmentation less balanced. DBSCAN with eps = 1.5 and

min_samples = 10 finds three dense clusters and marks about 0.50% of records as noise. These scores should be

compared with cluster sizes and business interpretation, not used in isolation.

DBSCAN parameter sensitivity

eps min_samples Clusters Noise % Silhouette (non-noise)

0.75 5 9 7.55 0.136

0.75 10 4 9.55 0.160

1.00 5 8 2.98 0.020

1.00 10 7 5.35 0.022

1.25 5 4 0.98 0.627

1.25 10 3 1.60 0.634

1.50 5 3 0.40 0.639

1.50 10 3 0.50 0.639

2.00 5 3 0.08 0.639

2.00 10 3 0.40 0.639

2.50 5 2 0.00 0.659

2.50 10 2 0.00 0.659

At eps = 0.75-1.00, DBSCAN generates more small clusters and/or more noise. Around eps = 1.25-2.00, it finds a small set of dense

groups and low noise. At eps = 2.50, the labels collapse to a two-group split. This illustrates why eps and min_samples must be

assessed jointly.

Cluster profile cross-check: Ward and DBSCAN

The alternative algorithms show a similar broad pattern: a core group, a smaller recent-flight-active group, and a

low-reward-activity group. DBSCAN also marks 20 records as noise; those records have unusually high median bonus

miles in this run, so they merit review rather than automatic deletion. Cluster IDs are algorithm-specific and must not

be compared as if they were the same labels.

Algorithm Profile Cluster n Median bonus miles Mean flight miles Award rate

Ward Core reward earners 0 3,311 9,680 438 39.2%

Ward Active flyers 1 226 11,875 1,720 64.6%

Ward Low reward activity 2 462 0 0 8.0%

DBSCAN Noise / unusual profiles Noise (-1) 20 62,914 788 50.0%

DBSCAN Core reward earners 0 3,761 6,654 384 35.3%

DBSCAN Active flyers 1 206 12,898 1,836 69.4%

DBSCAN Low reward activity 2 12 0 0 0.0%

4. Final K-Means segments and interpretation

The k = 3 K-Means solution is used for the descriptive profile below. Segment names are assigned after checking the

raw-data profiles; numerical cluster labels themselves are arbitrary.

Segment profile table

Segment Customers Median balance Median bonus miles Mean flight miles (12 mo.) Award rate

Core reward earners 2,857 56,756 13,495 501 43.3%

Low reward activity 916 14,079 0 20 10.7%

Active flyers 226 66,838 11,875 1,720 64.6%

Figure 3. Active flyers have the highest mean recent-flight mileage (about 1,720 miles) and the highest award rate (64.6%). Low

reward activity has no median bonus miles or bonus transactions and the lowest award rate (10.7%). Core reward earners form the

largest group (2,857 customers), with median bonus miles of 13,495 and an award rate of 43.3%.

Interpretation and possible actions

- Low reward activity (916; 22.9%): lower median balance, zero median bonus miles and bonus transactions, and

low redemption activity. Consider a low-cost re-engagement test, but first confirm whether these customers are still

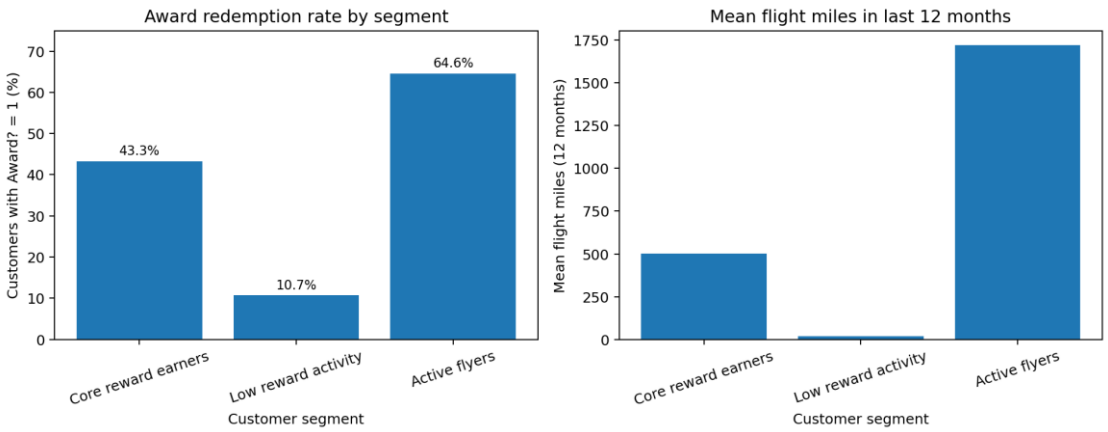
active or eligible.

- Core reward earners (2,857; 71.4%): most customers sit here, with a material reward balance and moderate recent

flight activity. Test personalized reward reminders or credit-card-linked earning offers.

- Active flyers (226; 5.7%): high mean recent flight mileage, a longer median enrollment duration, and high award

rate. Review retention and relevant tier/reward experiences for this group.



5. Visual separation, limitations, and conclusion

Figure 4. Projection of the scaled features onto two principal components, colored by the selected K-Means labels. The view is a 2D

summary of the full feature space, not a complete representation of all dimensions. Overlap in this projection does not by itself

invalidate the clustering.

Model evaluation conclusion

No one algorithm dominates on every criterion. K-Means k = 2 has the best silhouette in the tested k range (0.659),

while k = 3 sacrifices some separation for a more actionable three-segment interpretation. Ward hierarchical clustering

also gives a relatively balanced three-group solution. DBSCAN is useful for surfacing unusual records, but the results

depend on eps and min_samples.

Limitations

- There is no target label against which to validate customer segments; internal metrics measure geometric separation,

not profit or retention impact.

- Capping, log transformation, RobustScaler, and treating the credit-card mile levels as numeric all influence the

distance calculations.

- The snapshot lacks demographics, customer margin, purchase history beyond the supplied fields, and future churn

or retention labels.

- Award? was excluded from clustering and used only as a profiling variable; segment-level award rates remain

descriptive, not causal.

Conclusion

The supplied dataset supports a defensible exploratory segmentation into low reward activity, core reward earners, and active flyers. These labels can guide hypothesis-driven experiments, but the airline should test response, incremental revenue, retention, and fairness before operational deployment. Submission artifact reminder: the executable notebook and Python script are included in the accompanying package..

Execution url: https://colab.research.google.com/drive/1IgQoZ-8GhcdetvHEL5XyUTmR4KdNthjV?usp=sharing



INSTITUTE OF TECHNOLOGY
స్వయం తేజస్విన్ భవ
1979